

Synopsis

Project Title: Heart Disease Prediction

Problem Id: 13

Rishika Kapoor (Team Leader)

2028476

Section: U

Riddhima Gairola

2028473

Section: U

Aakshi

2028375

Section: U

• **Problem Description:**

Heart disease is one of the leading causes of death worldwide. Early detection and prediction of heart disease can help doctors take preventive measures and provide timely treatment to patients. However, analyzing large amounts of medical data manually can be difficult and time-consuming. Machine Learning techniques can help in identifying patterns in medical data and predicting whether a person is at risk of heart disease. The aim of this project is to design and develop an end-to-end Machine Learning classification system that predicts the presence of heart disease using clinical patient data. The system will use a heart disease dataset containing medical attributes such as age, blood pressure, cholesterol level, chest pain type, and heart rate. The dataset will be preprocessed by handling missing values, normalizing numerical data, and converting categorical variables into numerical form. Different Machine Learning models such as Logistic Regression, K-Nearest Neighbors (KNN), and Support Vector Machine (SVM) will be trained on the dataset. The performance of these models will be evaluated using metrics like accuracy, precision, recall, F1-score, confusion matrix, and ROC curve. The final model will help in predicting whether a patient has heart disease (1) or not (0).

• **Steps of Implementation:**

1. Data Collection:

A heart disease dataset is collected in CSV format containing patient medical attributes such as age, cholesterol, blood pressure, chest pain type, and heart rate.

2. Data Understanding:

- Checked dataset structure using `.info()`
- Identified data types (numerical and categorical)
- Observed dataset size and features

3. Data Cleaning:

- Checked and handled missing values
- Removed duplicate records (large number of duplicates found)
- Ensured dataset consistency

4. Handling Missing Value:

- Numerical columns filled using median (robust to outliers)
- Categorical columns filled using mode

5. Data Transformation:

- Converted categorical variables using one-hot encoding
- Converted data into numerical format for ML models

6. Outlier Detection and Removal:

- Applied IQR (Interquartile Range) method
- Removed extreme values to improve model accuracy

ML PIPELINE:

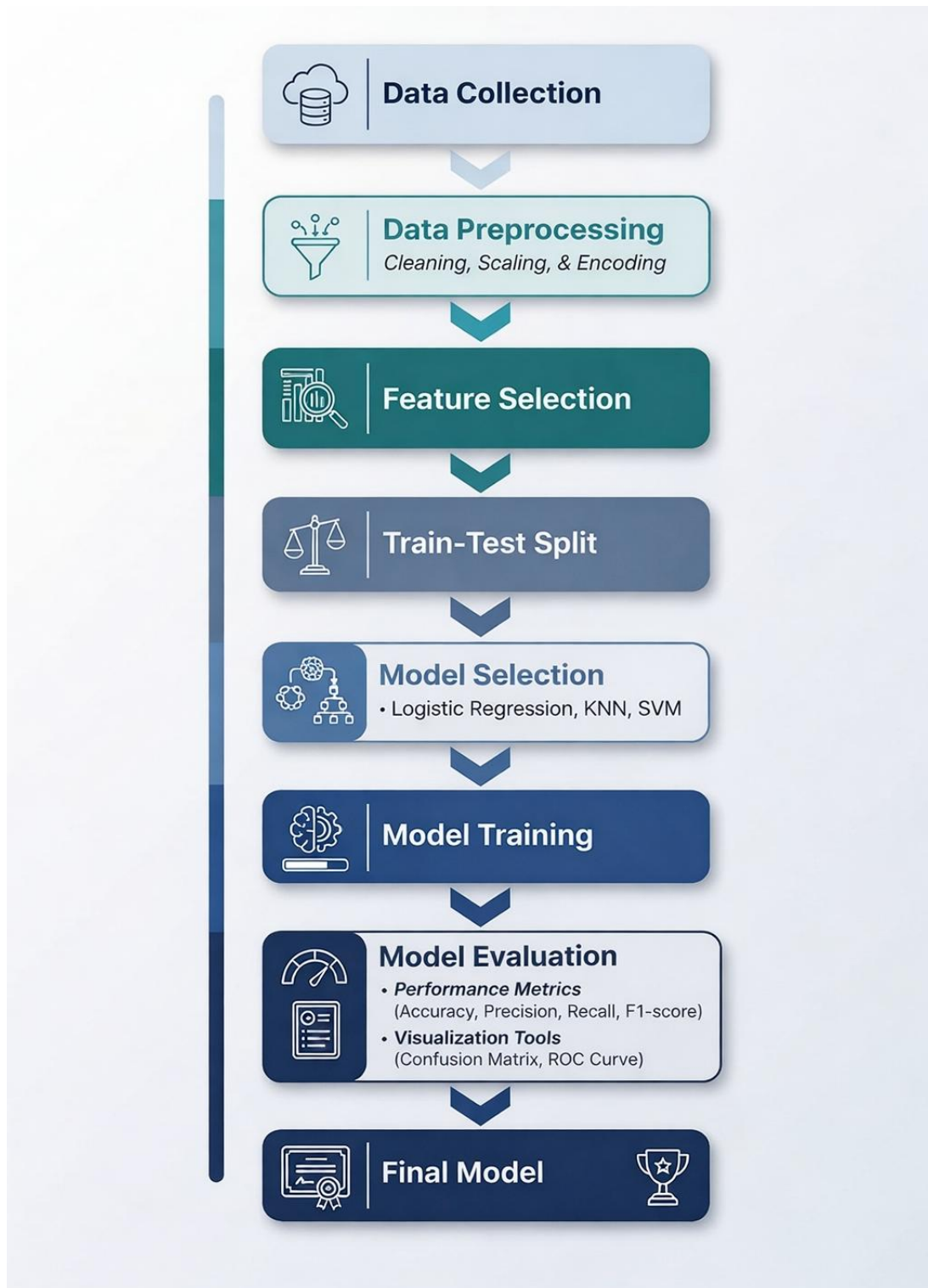


Fig. 1 Architecture for Heart Disease Prediction

- **Expected Modules/ Libraries to be used :**

1. **NumPy**:- Used for performing numerical computations and handling arrays efficiently.
2. **Pandas**:- Used for data manipulation, cleaning, and analysis of the dataset.
3. **Matplotlib**:- Used for plotting graphs and visualizing data distributions and model performance.
4. **Seaborn**:- Used for advanced statistical data visualization such as heatmaps and correlation plots.

- **Expected Evaluation metrics and Plots to be used:**

Not for PHASE 1.

- **Required topics from the subject:**

1. **Strings**:- Strings will be used for handling textual data, reading dataset column names, and displaying output messages in the program.
2. **Functions**:- Functions are used to organize the code into reusable blocks, making the implementation more structured and efficient.
3. **Files**:- File handling will be used to read the dataset file (CSV format) and store the data for processing and analysis.

- **Platform Required:**

The project will be implemented using Python programming language. Development and execution are done using platform Jupyter Notebook which provides an interactive environment for writing and testing machine learning code.

- **Books and Link Sources:**

1. GeeksforGeeks (ML Basics):
(<https://www.geeksforgeeks.org/data-analysis/exploratory-data-analysis-in-python/>)
2. W3Schools Python:
(https://www.w3schools.com/python/python_ml_linear_regression.asp)
3. Kaggle Dataset:
(<https://www.kaggle.com/datasets/mohamedkhairyelbanna/heartdiseasetrain-test>)
4. Class Notes